

Multiple Choice (10 marks)

1. In a laboratory procedure, a student observes different sizes of soil particles. Which tool is best for the investigation?
 - (a) a hand lens with 1 X magnification
 - (b) a microscope with 40 X magnification
 - (c) a light microscope with 100 X magnification
 - (d) a dissecting microscope with 4 X magnification
2. What process causes igneous rock to transform into metamorphic rock?
 - (a) deposition in water
 - (b) slow cooling and crystallization
 - (c) weathering and erosion
 - (d) exposure to heat and high pressure
3. How many basic units of information in a DNA molecule are required to encode a single amino acid?
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
4. When a button is pushed on a battery-operated clock, the time display is illuminated. What process has taken place?
 - (a) Pressure has been converted to light.
 - (b) Force has been converted to light.
 - (c) Closing an electric circuit has produced light.
 - (d) Opening an electric circuit has produced light.
5. A mouse is placed at one end of a cardboard maze that has a piece of cheese at the other end. The mouse smells the cheese and learns how to find it. The mouse is using an
 - (a) acquired trait to gain an inherited trait.
 - (b) inherited trait to gain an acquired trait.
 - (c) acquired trait to gain another acquired trait.
 - (d) inherited trait to gain another inherited trait.

True / False (10 marks)

Answer True or False

6. True or False: Conventional diesel fuel is a renewable fuel option for diesel engines.
7. True or False: Plant cells have the ability to convert sunlight into energy through the process of photosynthesis.
8. True or False: A force pushing a rolling ball in the same direction will cause it to stop moving altogether.
9. True or False: While sledding down a hill, both kinetic energy and potential energy increase simultaneously.
10. True or False: All light waves are transverse waves, meaning they oscillate perpendicular to the direction of wave propagation.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the atomic mass of an atom is determined, using an atom with 20 electrons, 21 neutrons, and 20 protons as an example.
12. Discuss and provide a detailed example of a physical change and a chemical change, explaining the differences between the two processes and their significance in everyday life.

End of Paper